

SF6 RING MAIN UNIT

PSEC

Metal-enclosed switchgear for secondary distribution networks
12 kV & 24 kV



630 A

RATED BUSBAR
CURRENT

25 kA

SHORT-TIME
WITHSTAND · 1 S

SF6

SEALED-FOR-LIFE
SWITCHING

IEC

62271 SERIES
COMPLIANT

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. We are not merely a contracting and supply firm — we are a fully integrated engineering partner specialising in the design and manufacture of Medium Voltage (MV) and Low Voltage (LV) power distribution systems, adhering to the most stringent international technical standards.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units and LV main and sub-main distribution boards are characterised by exceptional operational efficiency and a long life cycle. We bridge mechanical durability with digital intelligence, ready for integration with the latest smart-grid and control technologies.

Quality is not a department — it is our work culture



Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.



Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.



Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012

ESTABLISHED · CAIRO

2MANUFACTURING
FACILITIES**MV+LV**

INTEGRATED PORTFOLIO

IECCERTIFIED QUALITY
SYSTEM

“ Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. ”

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How to use this catalogue

Sections 07–09 give assembly-level data. Sections 11–15 detail the switching devices and components. Sections 16–19 support unit selection, configuration and installation planning.



PSEC MULTI-WAY ASSEMBLY — SF6 RING MAIN UNIT ON THE POWERLINE FACTORY FLOOR

PSEC AT A GLANCE

PSEC is a metal-enclosed Ring Main Unit for indoor secondary distribution at 12 kV and 24 kV. It combines an air-insulated, metal-enclosed cubicle system (AIS) with hermetically sealed SF6 three-position load break switches — a proven architecture that is compact, climate-independent and maintenance-free.

- Metal-enclosed unit for indoor installation, air-insulated busbar system
- Three-position load break switch: **Closed – Open – Earthed**, in a sealed SF6 enclosure
- Fully independent of the environment — dust, small insects, moisture and climate
- Sealed-for-life switch poles: no gas refilling throughout the service life
- All operations from the front; automatic position-driven interlocking protects the operator
- Fuse striker pin trips the switch mechanism automatically on fuse operation
- Flexible platform: hosts ABB GSec (main solution) or Murge MXGN switch-disconnectors



PSEC ring main unit — typical 3-way configuration

12 / 24 kV

RATED VOLTAGE

630 A

MAIN BUSBAR

25 kA

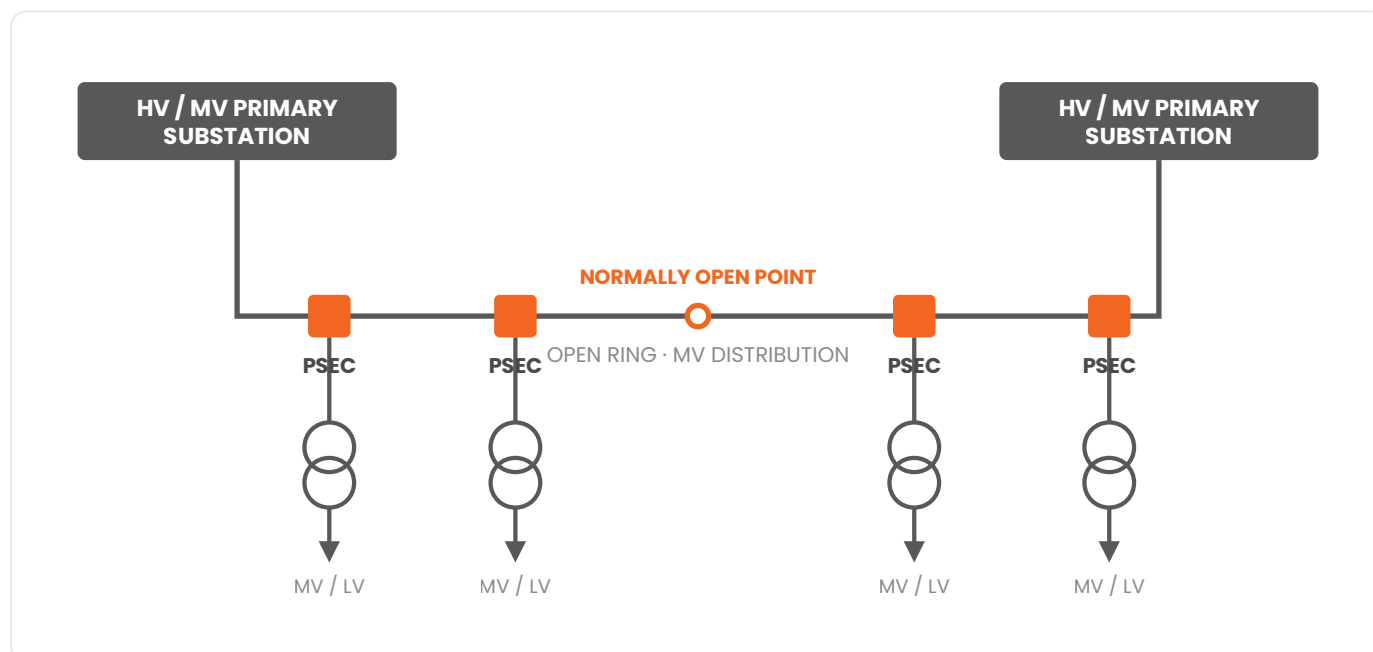
SHORT-TIME · 1 S

0SCHEDULED
MAINTENANCE

Designation. Functional units are combined to build the required scheme: **PCC** Powerline cable cubicle · **PFC** Powerline fuse cubicle · **PMC** metering cubicle. See Functional Units, page 23.

FIELD OF APPLICATION

PSEC connects, supplies and protects distribution transformers and cable feeders on open-ring or radial MV networks. Ring operation allows every substation to be fed from two directions — faults are isolated locally and supply is restored quickly, keeping service continuity high.



Typical open-ring secondary distribution network with PSEC units at each MV/LV substation

Typical duties

- Ring main and radial network switching (incoming / outgoing feeders)
- Distribution transformer protection by switch-fuse combination or downstream unit
- Network sectionalising and load transfer
- Utility metering points with dedicated measurement cubicles

Typical environments

- Utility distribution networks
- Compact & package substations
- Residential compounds
- Commercial & mixed-use buildings
- Industrial & manufacturing plants
- Infrastructure projects
- Power & transformer stations
- Mining applications

SAFETY BY DESIGN

Every switching operation on PSEC is designed to be safe for personnel – even with power present in the cables. Interlocks operate automatically according to the switch position, so the safe sequence is enforced by the equipment itself, not by procedure alone.



Safe with live cables

The unit can be approached and operated safely while cables are energised. Live parts are metal-enclosed and segregated from the operator.



Automatic interlocking

Interlocks driven by the switch position prevent any incorrect operating sequence. Operation is only possible with the door fully closed.



Earthed fuse access

The fuse compartment can only be accessed once the load break switch is in the earthed position – no exposure to live parts.



Clear status indication

A mimic diagram on the front fascia gives a trusted, unambiguous indication of the switching state: Closed, Open or Earthed.



Voltage detection

A capacitive voltage indicator (IEC 61958) confirms whether the connected cables are live before any operation or earthing.



Fail-safe fuse tripping

On fuse operation, the striker pin trips the switch mechanism automatically, disconnecting all three phases of the faulted circuit.

IAC · AF

Internal arc classification with front accessibility (AF) and IP3X protection on the front face: in the unlikely event of an internal arc, the operator standing at the front of the unit is protected. Enclosure finished in RAL 7035.

PRODUCT FEATURES



Technology

- Metal-enclosed unit for indoor installation
- Air-insulated cubicle system (AIS) with insulated busbars
- SF6 three-position load break switch: Closed – Open – Earthed
- Easy installation, maintenance-free operation
- Independent of climate and installation environment



Durability

- Switch tank hermetically sealed for life
- Unaffected by dirt, small insects and moisture
- Load break switch operated from the front of the RMU
- Automatic position-based interlocking system
- No gas recharging needed throughout the service life



Safety

- Safe operation with power present in cables
- Mimic diagram with clear operation status
- Operation only possible with the door closed
- Fuse compartment accessible only when earthed
- Voltage detector to verify live cables



Cost saving

- No maintenance other than HRC fuse replacement after operation
- Compact design – minimum space to install and operate
- Long service life with no replacement costs
- One platform for network, protection and metering functions
- Local manufacture: fast delivery and lifetime support

Compact by design

A complete 3-way ring main unit – two incoming ring switches and one transformer protection way – occupies a footprint of roughly **1.55 × 1.07 m**. Where space is the constraint, PSEC is the answer: substations, basements, kiosks and package substations.

554

 MM · WAY
WIDTH

1070

MM · DEPTH

1700

MM · HEIGHT

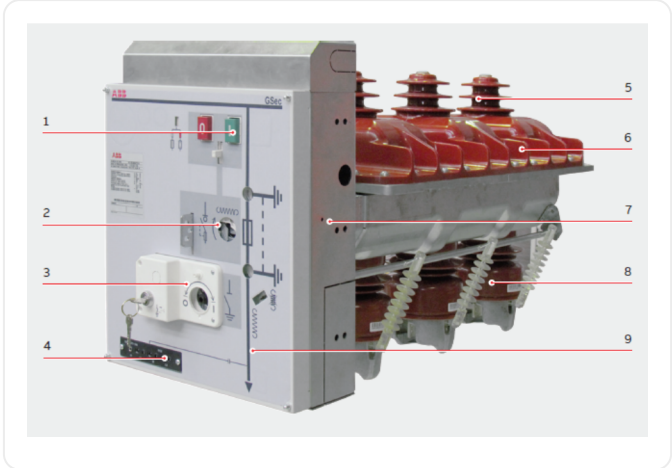
CONSTRUCTION & DESIGN

Each PSEC functional unit is a free-standing, metal-enclosed cubicle (LSC2A partition class) built around four clearly segregated zones. Cubicles bolt together into multi-way assemblies sharing a common insulated busbar system.

Compartment	Description
Busbar zone	Air-insulated 630 A main busbars, fully insulated (taped/sleeved) for maximum reliability
Switching device	Three-position SF6 switch-disconnector in a hermetically sealed stainless-steel enclosure
Cable compartment	Front-access cable termination for single-core cables up to 300 mm², with capacitive voltage indication, earthing and optional heater
LV compartment	Secondary wiring, auxiliary devices, indication and control accessories (230 V AC)

Enclosure

- Sheet-steel enclosure, powder coated RAL 7035
- IP3X on the front face · internal arc classification AF
- Mimic diagram, padlocking and key-locking facilities
- Anti-condensation heater and internal light options



1	Opening / closing push-buttons	6	Enclosure (power part)
2	Line operating lever seat	7	Operating mechanism housing
3	Earth operating lever seat	8	Bottom insulators
4	Voltage indicators	9	Mimic diagram
5	Top insulators		

Heart of the unit: the three-position SF6 switch-disconnector module

MAIN CHARACTERISTICS

Ratings of the complete PSEC assembly. Component-level data for the installed switch-disconnector is given in sections 11 and 12.

Characteristic	Unit	PSEC 12	PSEC 24
Rated voltage	kV	12	24
Service voltage	kV	11	22
Rated frequency	Hz	50	50
Rated power-frequency withstand voltage (1 min)	kV	28	50
Rated lightning impulse withstand voltage (1.2/50 μ s)	kV	75	125
Rated current, main busbars	A	630	630
Rated short-time withstand current (1 s) ⁽¹⁾	kA	25	20
Rated peak withstand current ⁽¹⁾	kA	63	50
Internal arc classification	—	IAC · AF · front accessibility	
Degree of protection, front face	—	IP3X	
Partition class	—	LSC2A	
Installation	—	Indoor	
Enclosure colour	—	RAL 7035	

(1) Values according to the selected load break switch type and duty. ABB GSec supports up to 25 kA (2 s) at both voltage levels — refer to the switch characteristics on pages 16 and 19 for device-level ratings.

Voltage detection

Capacitive voltage presence indication system (VPIS) to IEC 61958 fitted as standard on cable ways.

Earth fault indication

Earth fault indicator supplied on ring switch cubicles for fast localisation of cable faults.

Cable connection

Front cable termination, single-core cables up to 300 mm² (switch ways) / 95 mm² (fuse ways).

STANDARDS & SERVICE CONDITIONS

PSEC is designed, manufactured and routine-tested in accordance with the IEC 62271 series.

Standard	Scope
IEC 62271-1	High-voltage switchgear and controlgear — Part 1: Common specifications
IEC 62271-100	Part 100: Alternating-current circuit-breakers
IEC 62271-102	Part 102: Alternating-current disconnectors and earthing switches
IEC 62271-103	Part 103: Switches for rated voltages above 1 kV up to and including 52 kV
IEC 62271-105	Part 105: Alternating-current switch-fuse combinations
IEC 62271-200	Part 200: AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV up to and including 52 kV
IEC 61958	Voltage presence indicating systems (VPIS)

Environmental & additional data

Condition	Specification
Operating temperature	-5 °C to +40 °C
Storage temperature	-5 °C minimum
Installation altitude	Up to 1,000 m above sea level
Relative humidity	Max. 95 %
Protection index	IP3X on front face
Internal arc classification	AF — front accessibility
Switchgear colour	RAL 7035
Busbar treatment	Insulated
Packing	Domestic
Factory acceptance test	Powerline standard FAT (one day)

Why it matters

- **Type-conformant design** — ratings verified against the IEC 62271 series requirements for metal-enclosed switchgear.
- **Routine tests on every unit** — dielectric, mechanical operation and auxiliary circuit checks before dispatch.
- **Witnessed FAT** — a structured one-day factory acceptance test programme open to client witnessing.

For installations above 1,000 m or outside the standard temperature range, contact Powerline engineering — the switch-disconnector itself is qualified up to 4,000 m, allowing project-specific derating studies.

CONTROL & WIRING CIRCUITS

Item	Specification
Local control voltage	230 V AC
Auxiliary voltage, ES electromechanical lock	230 V AC
Control circuit cross-section	1 mm ²
Voltage circuit cross-section	1.5 mm ²
Current circuit cross-section	2.5 mm ²
Earthing circuit cross-section	2.5 mm ²
Interconnection circuit cross-section	2.5 mm ²
Wiring cable type	Standard PVC

Wiring colour code	Colour
Auxiliary AC circuits	Black
Auxiliary DC circuits	Black
CT circuits	Black
VT circuits	Black
Earthing circuits	Yellow / Green

LV compartment & secondary equipment

Each cubicle carries a low-voltage compartment for secondary devices, factory-wired to terminal blocks and ready for site connection:

- Voltage presence indicator with fixed lamps (VPIS)
- Earth fault indicator on ring switch units
- Anti-condensation heater, 230 V AC, in the cable compartment
- Internal lighting for the cable compartment
- Fuse status indication and fuse-blown signalling contacts (fuse ways)
- Auxiliary contacts for remote position indication
- Provision for motor operation and shunt releases (options)

Smart-grid ready

With motorised mechanisms, auxiliary contacts and fault passage indication, PSEC integrates readily with RTUs and SCADA for remote monitoring and network automation.

SECTION 10

Switching devices

PSEC is engineered as an open platform: the same cubicle, busbar and interlocking system hosts more than one make of three-position SF6 switch-disconnector, giving specifiers flexibility of supply without changing the substation layout.

ABB GSec · main solution

Three-position SF6 switch-disconnector — pages 13–17

Murge MXGN · alternative

Three-position SF6 load break switch — pages 18–19



THREE-POSITION SF6 SWITCH-
DISCONNECTOR

3

POSITIONS: C / O / E

SF6

SEALED FOR LIFE

1S / 2S

SPRING MECHANISMS

IP4X

SWITCH ENCLOSURE

ABB GSEC LOAD BREAK SWITCH

GSec is a three-position SF6 gas-insulated switch-disconnector designed for medium voltage secondary distribution switchgear — the main switching solution installed in PSEC.

GSec is used in line incoming/outgoing panels and in combination with fuses. Typical applications are supplying lines, protecting transformers and switching in ring networks. The three positions — line closed, open, earth closed — are achieved by a single moving contact system, so the switch position is always unambiguous.

- Switching functions and earthing in one sealed SF6 enclosure
- Operating speeds independent of the operator (stored-energy springs)
- Making capacity on the earthing function for fault-making duty
- Front-operated: line and earth lever seats plus push-buttons
- Gas presence indicator option for continuous verification
- Motor operation option for remote control and automation

Three variants, one platform

GSec/T1 single-spring mechanism for ring ways · **GSec/T2** double-spring stored energy mechanism · **GSec/T2F** double-spring for switch-fuse combinations.



GSec three-position SF6 switch-disconnector

Position sequence

Closed → Open → Earthed. Movement between line and earth seats always passes through the open position — an inherent, mechanical guarantee of the safe sequence.

GSEC SELECTION & TYPES

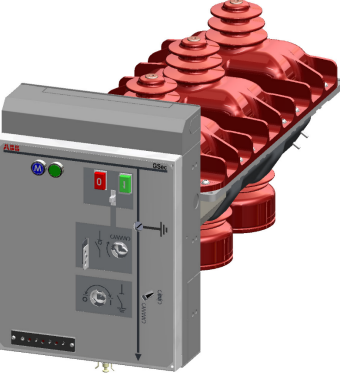
All three variants share the same sealed SF6 pole and three-position architecture; they differ in the operating mechanism and fuse-tripping features.



GSec / T1

1S — single-spring operating mechanism.

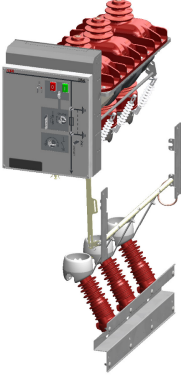
Opening and closing at speeds independent of the operator. The standard choice for incoming / outgoing ring units (PCC).



GSec / T2

2S — double-spring stored-energy mechanism.

Stored-energy opening and closing, independent of the operator. For incoming / outgoing units where remote or rapid switching is required.



GSec / T2F

2S — double-spring stored-energy mechanism.

Version for switch-fuse combinations (PFC): fuse striker tripping, blown-fuse indication and downstream earthing switch.

Standard supply	GSec/T1	GSec/T2	GSec/T2F
Switch-disconnector	●	●	●
Operating mechanism	1S single spring	2S double spring	2S double spring
Door interlock & operating lever	●	●	●
Fuses-blown indicator & tripping system	—	—	●
Separate earthing switch 5 kAp & fuse holder (supplied loose)	—	—	●

GSEC OPTIONS & ACCESSORIES

Factory-fitted optional accessories extend each GSec variant for automation, supervision and key-managed safety schemes.

Optional accessory	GSec/T1	GSec/T2	GSec/T2F
Auxiliary contacts	●	●	●
Motor for the operating mechanism	●	●	●
VPIS voltage indicators	●	●	●
Key lock for line seat	●	●	●
Key lock for earth seat	●	●	●
Coil preventing lever insertion in earth seat	●	●	●
Coil preventing lever insertion in line seat	●	—	—
Shunt opening release	—	●	●
Shunt closing release	—	●	●
Gas presence indicators	●	●	●
Fuses-blown signalling contact	—	—	●



Motorisation

Motor operators allow remote opening/closing and integration into automatic transfer or network automation schemes without changing the cubicle.



Key-managed safety

Line and earth seat key locks plus lever-blocking coils enforce switching programmes across several substations — the right sequence, every time.



Supervision

Auxiliary contacts, gas presence indication and fuse-blown signalling keep the control room informed of every switch position and event.

GSEC ELECTRICAL CHARACTERISTICS

One insulation platform, two operating mechanisms: insulation values are common to all variants, while current-dependent duties differ between the 1S and 2S mechanisms.

Characteristic	Unit	GSec/T1 12 kV	GSec/T1 24 kV	T2 · T2F 12 kV	T2 · T2F 24 kV
Rated voltage	kV	12	24	12	24
Power-frequency withstand (1 min) – to earth	kV	28	50	28	50
Power-frequency withstand – across open contacts	kV	32	60	32	60
Lightning impulse withstand (1.2/50 μs) – to earth	kV	75	125	75	125
Lightning impulse withstand – across open contacts	kV	85	145	85	145
Rated frequency	Hz	50 / 60			
Rated current (40 °C)	A	800	630	800	630
Rated short-time withstand current	kA	21 (3 s) / 25 (2 s)			
Making capacity (peak)	kAp	65	54.5	41.5	41.5
Making capacity, separate earthing switch	kAp	62.5 (GSec/IB)		5 (GSec/T2F)	
Breaking capacity					
– Active load	A	800	630	630	630
– Ring circuits	A	800	630	630	630
– No-load transformers	A	16			
– No-load lines	A	25			
– No-load cables	A	50			

Data of the switch-disconnector device according to the manufacturer. Ratings of the complete PSEC functional unit are given on page 09.

1S – single-spring mechanism (T1)

Charged and released in one movement of the operating lever; opening and closing speeds are independent of the operator. The workhorse for ring and network switching ways.

2S – double-spring stored energy (T2 · T2F)

Energy is stored in a separate spring and released on command – enabling shunt releases, motorisation and fuse-striker tripping on the T2F for switch-fuse combinations.

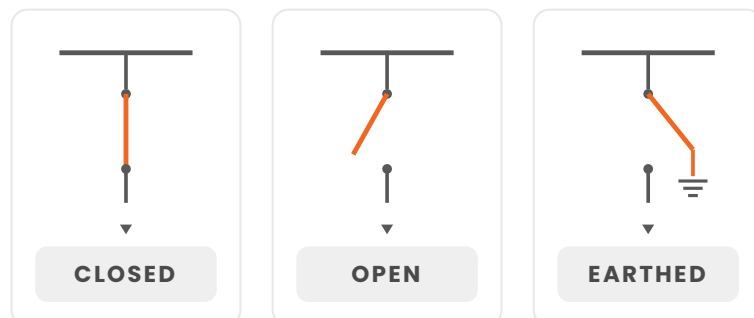
GSEC GENERAL DATA & POSITIONS

Other characteristics	Unit	Value
Distance between phases	mm	230
SF6 gauge pressure (filling)	bar	0.48
Weight of SF6 contained	kg	0.21
Internal volume	litres	25
IP protection class (switch enclosure)	—	IP2X
Operating temperature	°C	-5 ... +40
Storage temperature	°C	-25 ... +40
Maximum installation altitude	m	4,000

Minimal gas, sealed for life

Only 0.21 kg of SF6 at a low filling pressure of 0.48 bar — a sealed pressure system requiring no gas handling on site, for the entire service life of the switch.

Switching positions of the load break switch



- One moving contact system — the position indication is inherently true
- Transition line ↔ earth always passes through **Open**
- Earthing function with full making capacity for safe fault-making
- Padlock and key-lock provisions on both lever seats

MURGE MXGN LOAD BREAK SWITCH

In early 2024, Powerline's R&D team upgraded the PSEC platform to host multiple load break switch types. Among the most successful integrations is the MXGN switch by Murge.

Murge is a leading Chinese manufacturer of electrical solutions, specialising in gas-insulated switchgear and MV & LV components and accessories. The co-operation between Powerline and Murge — integrating the MXGN load break switch into the PSEC enclosure — has produced a unit that is highly competitive on both the technical and the commercial level.

Product overview

The MXGN is a fixed, metal-enclosed three-position switchgear device applied in 12 kV and 24 kV ring main units. It can also serve terminal power supplies or be fitted into cubicle-type substations for the control and protection of electric power — in power stations, transformer substations and distribution substations across industrial, manufacturing, residential and mining applications.

- Three positions — Closed / Open / Earthed — in a sealed SF6 enclosure
- Mechanical endurance class M2 (5,000 operations), electrical class E3
- Internal arc classified A-FLR · IP4X switch enclosure
- Same PSEC cubicle, busbar system, interlocks and accessories



MXGN three-position SF6 load break switch

Why a second source?

Dual sourcing of the switching device shortens delivery programmes and gives clients a commercially optimised alternative — with no change to the substation footprint, civil works or operating procedures.

Configurations

MXGN-equipped PSEC units are available in the same PCC, PFC and PMC cubicles — see pages 24–26.

MXGN TECHNICAL PARAMETERS

No.	Description	Unit	12 kV	24 kV
1	Rated voltage (Ur)	kV	12	24
2	Rated current (Ir)	A	630	630
3	Rated frequency (fr)	Hz	50	50
4	Rated power-frequency withstand voltage / 1 min (phase-to-phase & to-earth / across isolating distance)	kV	28 / 32	50 / 60
5	Rated lightning impulse withstand voltage (phase-to-phase & to-earth / across isolating distance)	kV	75 / 85	125 / 145
6	Rated short-time withstand current	kA / s	25 / 3	20 / 3
7	Rated peak withstand current	kA	62.5	50
8	Rated short-circuit making current	kA	62.5	50
9	Rated transfer breaking current	A	1,700	1,400
10	Rated closed-loop breaking current	A	630	630
11	Maximum rated current of the fuse	A	200	200
12	Mechanical endurance — main switch	ops	M2 (5,000)	M2 (5,000)
	Mechanical endurance — earthing switch	ops	M1 (2,000)	M1 (2,000)
13	Electrical endurance	—	Class E3	Class E3
14	Internal arc classification (IAC)	—	A-FLR 20 kA/1s · 25 kA/1s	A-FLR 16 kA/1s · 20 kA/1s
15	Degree of protection	—	IP4X	IP4X

Data of the switch device according to the manufacturer's type tests.

INSTRUMENT TRANSFORMERS

Current transformers

Ring-core type for protection and indication on cable ways; cast-epoxy DIN types for the metering cubicle.

Characteristic	Unit	Value
Max. system voltage	kV	0.6
Primary current	A	7.2 – 230.4
Secondary current	A	0.075
Rated burden	VA	0.1
Accuracy class	—	10P80
Short-time current	kA / 1 s	20
Rated frequency	Hz	50 / 60

Metering duty (PMC): cast-epoxy DIN CTs, primary per client requirement, core 1 A · 10 VA · class 0.5 · FS5 · 50 Hz.

Voltage transformers

Single-pole (phase-to-earth) epoxy-resin VTs for measurement and metering, with or without protection fuse.

Characteristic	Unit	Value
Highest voltage for equipment	kV	3.6 – 40.5
Primary voltage	V	3000/√3 – 35000/√3
Secondary voltage	V	100/√3 – 120/√3 · 100/3 – 120/3
Rated frequency	Hz	50
Fusing	—	With / without fuse

Metering duty (PMC): phase-earth VTs, core Us = 0.11/√3 kV · 50–100 VA · class 0.5.

Selection note. Instrument transformer ratios and burdens are engineered per project — primary currents, cores and accuracy classes are matched to the utility metering code and protection scheme. Powerline adjusts burdens accordingly at order stage.

OPTIONS & ACCESSORIES

Every PSEC way can be tailored with supervision, comfort and safety accessories — factory fitted and tested.



Voltage presence indicator (VPIS)

Push-button LED voltage indicator to IEC 61958, supplied by bushing-type capacitive dividers, confirming whether cables are live.



Earth fault indicator

Fault passage indication on ring ways for rapid localisation of cable faults and shorter restoration times.



Anti-condensation heater

230 V AC heater in the cable compartment prevents condensation in humid or cyclic-temperature environments.



Internal lighting

Cable compartment illumination for safe termination, inspection and maintenance work.



Locking schemes

Padlocks, key locks on line and earth seats, and 230 V AC electromechanical locking for switching-programme enforcement.



Motor operation

Motorised mechanisms for GSec/ TI, T2 and T2F allow remote switching and automation readiness.



Auxiliary contacts

Position signalling of switch and earthing switch for interlocking, SCADA and remote indication.



Gas presence indicator

Continuous indication of SF6 presence in the switch enclosure for asset supervision.



Fuse status signalling

Blown-fuse indicator and signalling contact on switch-fuse ways, with automatic three-phase tripping.

HRC POWER FUSES

PSEC switch-fuse ways (PFC) carry three HV HRC current-limiting fuses. The HRC power fuse interrupts high fault currents **before the first peak**, dramatically reducing the withstand duty imposed on the protected transformer and the associated equipment.

- Current-limiting interruption — cuts the fault before peak value
- High breaking capacity in a compact, enclosed design suitable for in-panel use
- Striker pin operates the trip indicator and trips the switch mechanism, disconnecting all three phases
- Standard equipment: 24 kV fuses up to 80 A (per project selection)
- Combination behaviour per IEC 62271-105

Coordination made simple

Fuse-link ratings below follow the fuse manufacturer's selection tables for distribution transformers without secondary-side coordination constraints. Final selection considers inrush, cyclic loading and utility rules — Powerline engineering validates every order. Maximum fuse rating accepted by the switch: 200 A.

Recommended fuse-link ratings [A] vs. transformer rating [kVA]

Service voltage	25	50	75	100	125	160	200	250	315	400	500	630	800	1000	1250	1600
11 / 12 kV	6	6	10	16	16	16	20	20	25	40	40	50	63	80	100	125
22 / 24 kV	6	6	6	6	10	10	16	16	16	20	20	25	40	40	50	63

Indicative values based on the fuse manufacturer's tables (CEF-type, back-up class); ratings above 80 A on request. Confirm final selection with Powerline engineering per IEC 62271-105 and the specific fuse type-test data.

< ½ cycle

CURRENT-LIMITING ACTION

80 A

STANDARD MAX. FITTED RATING

200 A

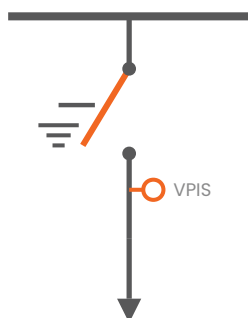
MAX. ACCEPTED BY SWITCH

3 φ

TRIPPING VIA STRIKER PIN

FUNCTIONAL UNITS

Three standardised cubicles combine into any secondary distribution scheme – from a single switch way to multi-way substations with transformer protection and utility metering.

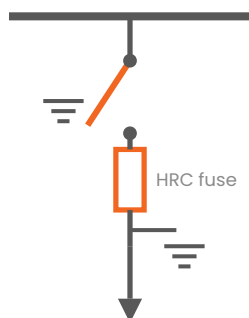


Cable up to 300 mm²

PCC

Powerline Cable Cubicle · 630 A.

Ring / network switching way with three-position switch-disconnector and earthing function.

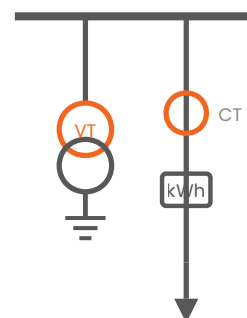


Cable up to 95 mm²

PFC

Powerline Fuse Cubicle · 630 A.

Transformer protection way with HRC fuses, striker tripping and downstream earthing.



Metering way

PMC

Powerline measurement cubicle.

Universal metering panel with VTs, CTs and utility metering provisions (LSC2A).

Model designation



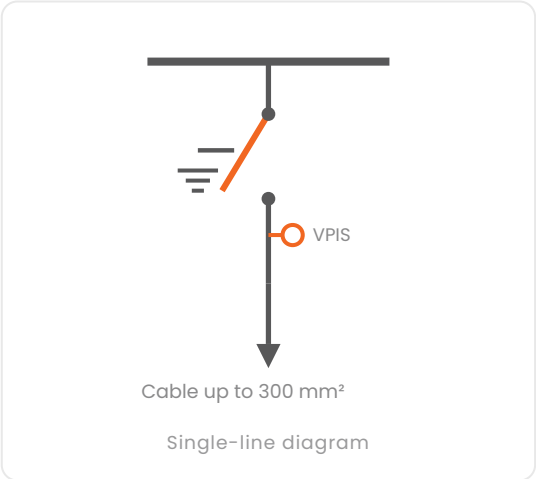
N = number of network LBS ways (PCC) · **F** = number of switch-fuse ways (PFC) · **M** = metering cubicle (PMC).

Example: **N2F1M** – two ring switch ways + one transformer protection way + metering: the classic compact substation.

PCC · POWERLINE CABLE CUBICLE

Ring and network switching way, rated 630 A. Metal-enclosed cubicle, LSC2A, hosting a three-position SF6 switch-disconnector with earthing function.

Qty	Composition
1	SF6 load break switch disconnector unit, cubicle type LSC2A
1	Three-position switch-disconnector with earthing switch and single-spring (1S) operating device — ABB GSec or Murge MXGN
1	Cable entry for 1-phase (single-core) cables up to 300 mm²
1	Voltage presence indicator with fixed lamps, type VPIS
1	Earth fault indicator
1	Anti-condensation heater, 230 V AC, cable compartment
1	Internal light
1	Insulated busbar system
1	Low-voltage compartment



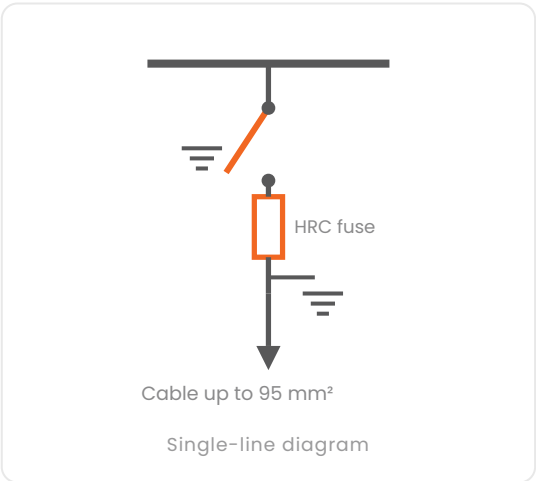
Dimensions	
Width	554 mm
Depth	1,070 mm
Height	1,700 mm

Options: motor operation, key locking schemes, auxiliary contacts, gas presence indicator.

PFC · POWERLINE FUSE CUBICLE

Transformer protection way, rated 630 A. Switch-fuse combination per IEC 62271-105 with automatic three-phase tripping on fuse operation and earthed-access fuse compartment.

Qty	Composition
1	SF6 load break switch with fuse combination unit, cubicle type LSC2A
1	Three-position switch-disconnector with earthing switch and double-spring (2S) stored-energy operating device — ABB GSec/T2F or Murge MXGN
3	HV HRC fuses, 24 kV, up to 80 A (P&C or equivalent), with striker pin
1	Fuse base and holder · fuse tripping system · fuse status indication
1	Downstream earthing switch with limited making capacity (2 kA)
1	Cable entry for 1-phase (single-core) cables up to 95 mm²
1	Voltage presence indicator with fixed lamps, type VPIS
1	Operating handle for switch-disconnector and earthing switch
1	Anti-condensation heater, 230 V AC · internal light
1	Insulated busbar system · low-voltage compartment



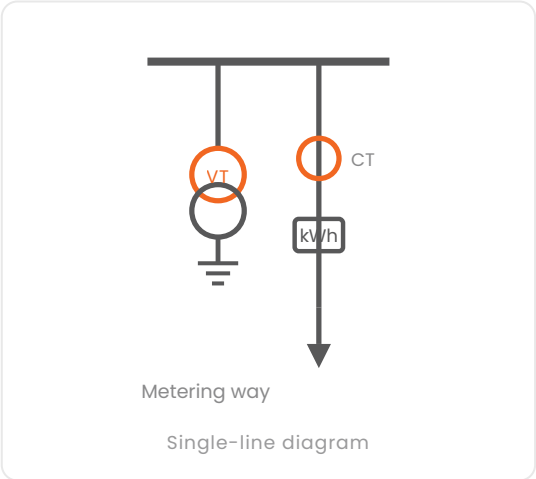
Dimensions	
Width	554 mm
Depth	1,070 mm
Height	1,700 mm

Fuse compartment access only with the way earthed — enforced by the interlocking system.

PMC · MEASUREMENT CUBICLE

Universal metering panel for utility revenue metering and network measurement, fully integrated into the PSEC busbar system.

Qty	Composition
1	Universal metering panel, cubicle type LSC2A
3	Cast-epoxy DIN current transformers — primary current per client requirement; core 1 A · 10 VA · class 0.5 · FS5 · 50 Hz
3	Voltage transformers, phase-to-earth — core $U_s = 0.11/\sqrt{3}$ kV · 50–100 VA · class 0.5
3	Ammeters
1	Voltmeter with 7-position selector switch
1	Voltage presence indicator with fixed lamps, type VPIS
1	Sealable door lock (utility sealing)
1	Insulated busbar system
1	Low-voltage compartment



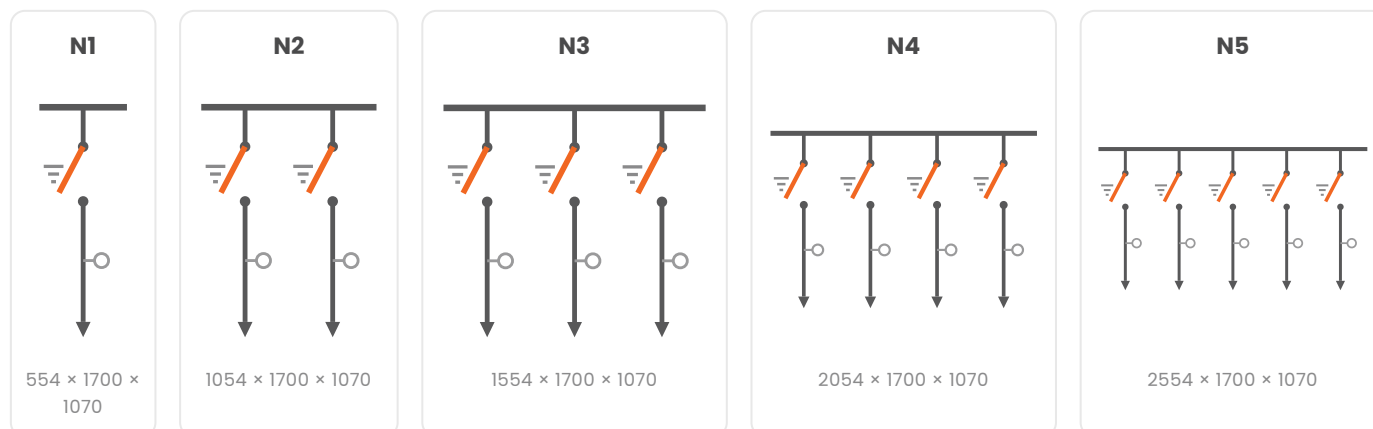
Dimensions	
Width (module, in assembly)	750 mm
Depth	1,070 mm
Height	1,700 mm

CT ratios and VT burdens are adjusted to the metering authority’s requirements at order stage.

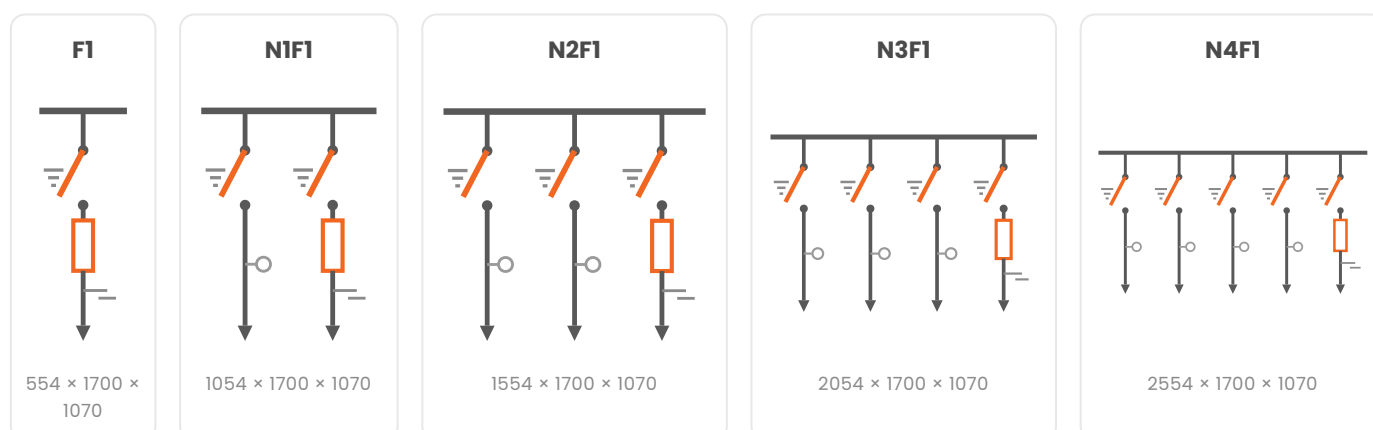
TYPES & DIAGRAMS

Standard PSEC assemblies with single-line diagrams. Model designation: **N** network switch way (PCC) · **F** switch-fuse way (PFC) · **M** metering cubicle (PMC). Dimensions W × H × D in mm.

1 Load break switch combinations — without metering



2 Transformer protection by fuses — without metering

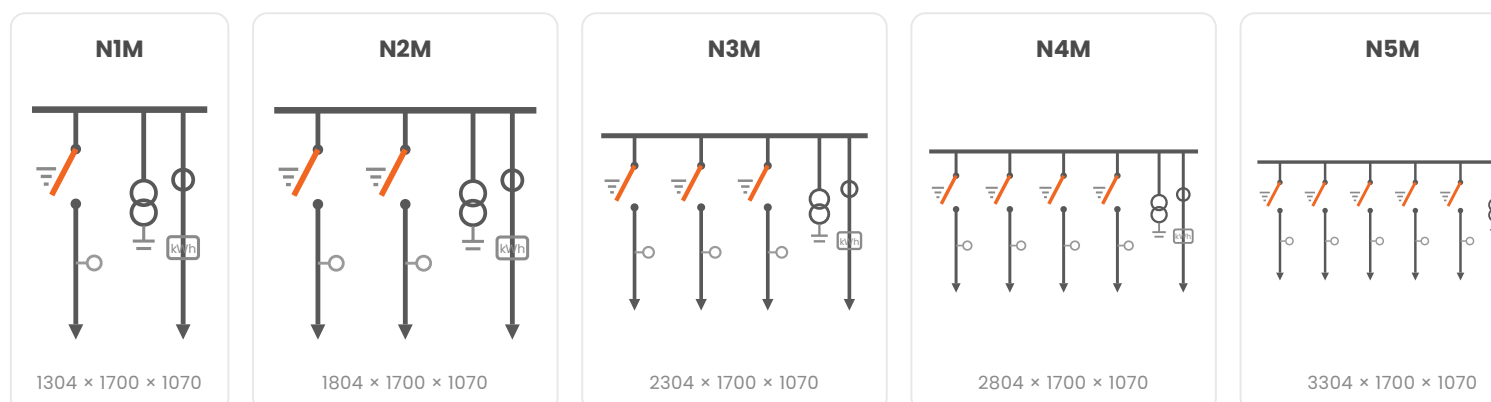


Width rule

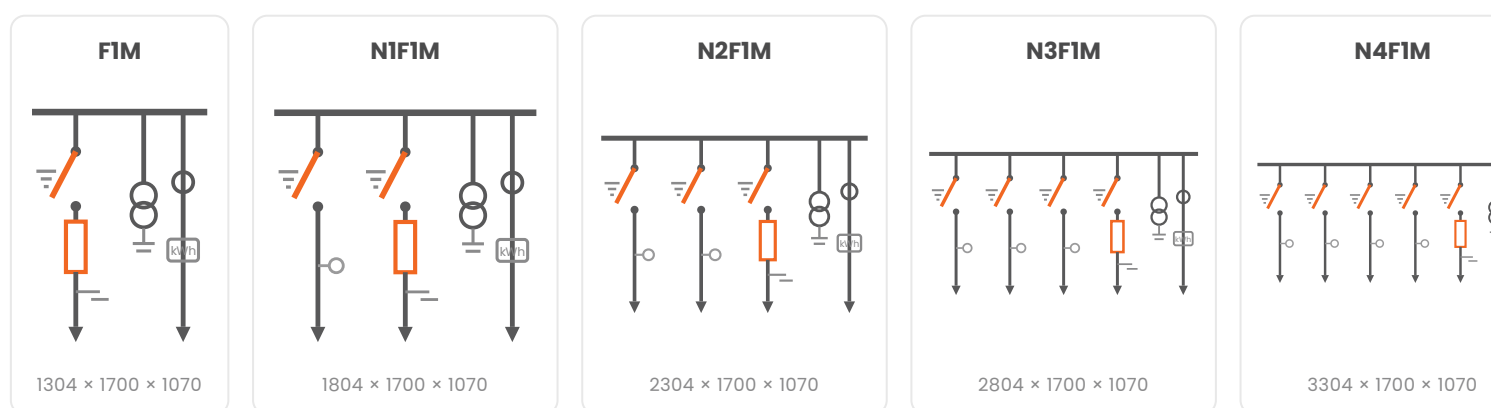
554 mm for the first switching way + 500 mm per additional N or F way + 750 mm for a metering module. All assemblies: height 1,700 mm · depth 1,070 mm.

COMBINATIONS WITH METERING

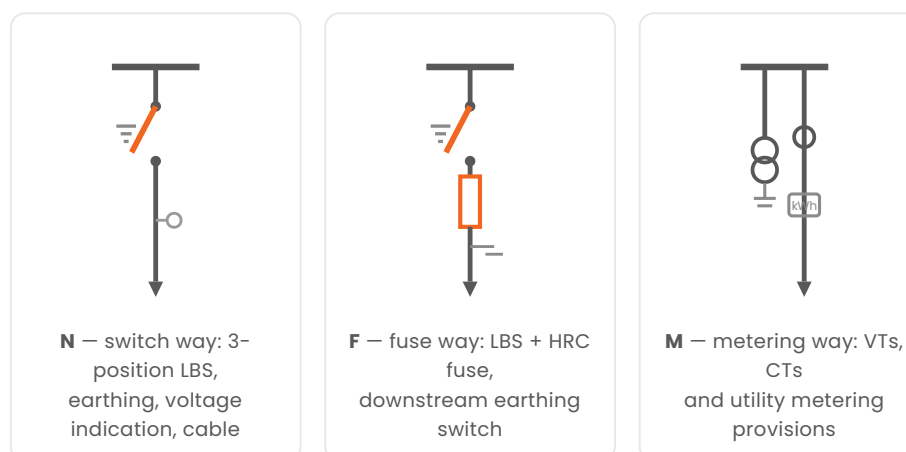
3 Load break switch combinations — with metering



4 Transformer protection by fuses — with metering



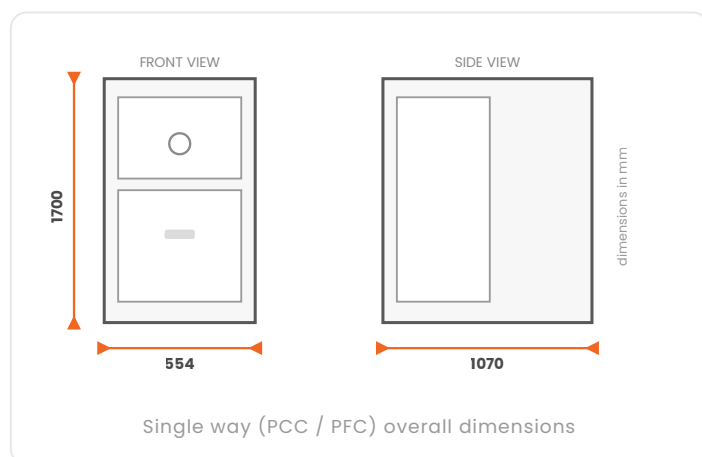
Symbol legend



Beyond the standard matrix

Larger schemes — up to five switching ways plus metering and multi-fuse arrangements — follow the same width rule. Every PSEC is assembled to the project single-line diagram; contact Powerline engineering for non-standard combinations.

INSTALLATION & SITE REQUIREMENTS



Module	Width	Height	Depth
PCC / PFC switching way	554 mm	1,700 mm	1,070 mm
Additional way (in assembly)	+ 500 mm	—	—
PMC metering module (in assembly)	+ 750 mm	—	—

Service conditions: -5...+40 °C · humidity max. 95 % · altitude up to 1,000 m (higher on request – switch device qualified to 4,000 m).

- **Indoor installation** on a level floor or plinth; the compact footprint suits substation rooms, basements and package substations
- **Front access only** required for operation, cable termination and fuse replacement – units can stand against the wall
- **Cable connection from the front**, single-core cables up to 300 mm² (switch ways) / 95 mm² (fuse ways); cable trench or duct below the unit recommended
- **Bolted assembly** of cubicles into multi-way units on site, or delivered as a complete factory-assembled RMU
- **Earthing** of the enclosure and earthing circuits per utility practice (2.5 mm² minimum earthing circuit section)
- **Easy handling** – compact, low-weight cubicles with lifting provisions

Commissioning & life

After cable termination and functional checks, PSEC is ready for service. No scheduled maintenance is required – only HRC fuse replacement after a fuse operation. The sealed switch needs no gas refilling for its entire service life.

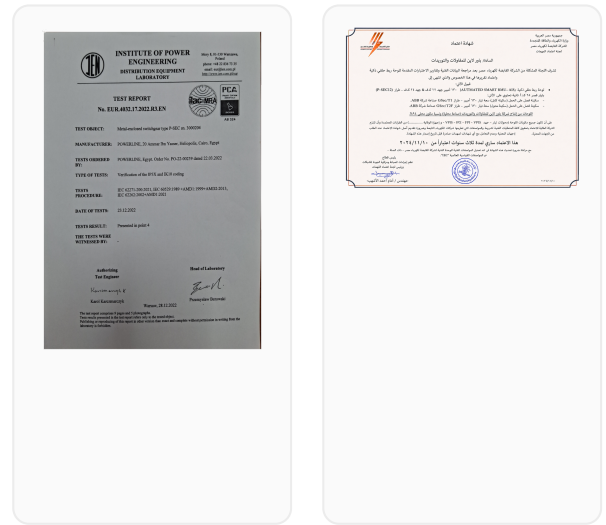
QUALITY, TESTING & CERTIFICATIONS

Routine tests — every unit

- Visual inspection, dimensional and build verification
- Power-frequency voltage test on the main circuit
- Voltage test on auxiliary and control circuits
- Mechanical operating tests and interlock verification
- Verification of correct wiring, indication and accessories
- Documented factory acceptance test (FAT) — one day, open to client witnessing

Type verification & approvals

- Design verified per IEC 62271-200 — verification of the IP3X protection and IK mechanical impact coding — witnessed by an independent international laboratory (Institute of Power Engineering, Warsaw)
- PSEC 12 kV & 24 kV approved and registered by the Egyptian electricity sector authorities, with GSec/T1 and GSec/T2F switch equipment and VPIS / EFI / FPI accessories, at $\geq 96\%$ local content
- Switch-disconnectors type-tested by their manufacturers (ABB / Murge) per IEC 62271-103 / -105



Independent laboratory test report (IEC 62271-200) · national approval certificate

Documentation with every delivery

Routine test certificates, as-built drawings, single-line diagram, operating manual and type-test evidence are supplied with each PSEC assembly.

Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

12 / 24

KV RATED VOLTAGE

630 A

MAIN BUSBAR

N • F • M

MODULAR FUNCTIONAL
UNITS

≥ 96 %

LOCAL CONTENT